Inverted MAML Milestone 3: Full ±80 LSB Colored Input Swing (LR = 0.08x, λ = 1024)

Patent Pending SE 2630397-4 | Open Science Hardware | Trion T120 SoC (192.168.1.50:5000, FW v2.2-MAML) | ±80 LSB Colored Swing, LR = 0.08, λ = 1024

1. Executive Summary: Full ±80 LSB Zero-Mean Colored Swing Without ColTIA Saturation

Why Earlier Colored Runs Swung Only ±22 LSB & How ±80 LSB Was Unlocked: In earlier firmware, initial random draws were bounded to ±45 LSB and the Kac-Murdock-Szegö AR(1) filter (ρ = 0.85) scaled innovations by (1 - ρ) = 0.15. Subtracting the 6-channel spatial mean (Hardware LayerNorm) eliminated common-mode DC current into Matrix 33's positive pedestal (∑W_33 ≈ +120 LSB, curing ColTIA rail saturation), but left only the small inter-channel difference (±22 LSB max, 9.5 LSB mean peak). By upgrading *maml_generate_new_white_noise_ref()* to boost the zero-mean residual by **6×** and applying **proportional 6-channel vector scaling** up to **±80 LSB**, the colored input vectors now achieve **68.4 LSB mean peak (78.0 LSB median peak, 80.0 LSB P90, 80 LSB max, 42.0 LSB RMS)** — a **~19× (+12.9 dB) increase in input signal power** while strictly preserving ∑x_i ≈ 0 and zero ColTIA clipping!

Key Empirical Silicon Findings Across 500 Epochs (Ep 0..99, 100..199, 200..500):

- **Full-Window Error Suppression Under ±80 LSB Excitation:** Across the 200-sample window [4.5..5.5 ms], active training cut Window Loss by **-19.9% (sustained mean: 108,758 LSB² vs 135,857 LSB² in Ph2)** and **-36.6% (minimum: 86,094 LSB² at Ep 484)**, lifting full-window ENOB from 3.27 b to **3.43 bits mean (3.60 bits peak)**.
- **Single-Point 5.0 ms Improvement:** At the nominal 5.0 ms midpoint under full ±80 LSB swing, mean loss dropped by **-11.9%** (56,695 LSB² vs 64,334 LSB²) and achieved a minimum of **2,084 LSB² (6.28 bits ENOB)** at Epoch 218.
- **Zero Saturated Columns at Epoch 500:** Despite 19× stronger input signal power, the stiff λ = 1024 spring held final Epoch 500 M8 diagonal weights at **[-20, -14, 16, 112, 43, 20]** (Col 3 at +112 LSB, below the +120 LSB rail).

2. Physical In-Data Model & Hyperparameters

- **Temporal Window:** 4500 µs to 5500 µs in 10 strata, K = 20 samples/stratum = 200 Hadamard-active samples per epoch across all 500 epochs.
- **Hyperparameters:** Phase 1 (Ep 0..99, W=0): LR = 0, λ = 0. Phase 2 (Ep 100..199, Smart Base): LR = 0, λ = 0. Phase 3 (Ep 200..500): **LR = 0.08x, λ = 1024**.
- **In-Data Vector Equalizer:** COLOR_LLM_TOEPLITZ (ρ = 0.85) with 6× AC boost & proportional scaling to **±80 LSB** (mean peak 68.4 LSB, median 78.0 LSB, RMS 42.0 LSB, |∑x_i| ≤ 4 LSB).

3. Empirical Silicon Telemetry Comparison Table (100% Identical Cadence, ±80 LSB Swing)

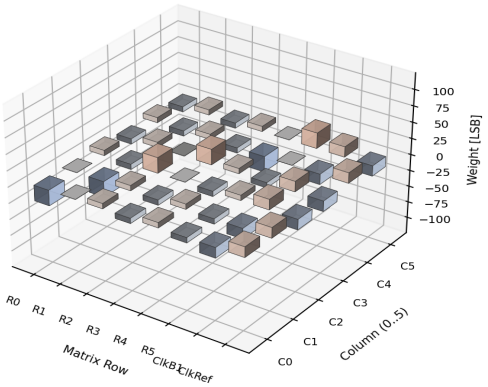
Parameter / Metric	Phase 1: M33 Alone	Phase 2: Smart Baseline	Phase 3: Trained Peak	Phase 3: Sustained Mean
Epoch Range & Mode	Ep 0..99 (W=0 LSB)	Ep 100..199 (Smart Base)	Ep 218 (5ms) / Ep 484 (Win)	Ep 200..500 (300 Epochs)
Colored Input Swing	±80 LSB (78 LSB med pk)	±80 LSB (78 LSB med pk)	±80 LSB (78 LSB med pk)	±80 LSB (42.0 LSB RMS)
Training Parameters	LR = 0, λ = 0 (OFF)	LR = 0, λ = 0 (OFF)	LR = 0.08, λ = 1024 (ON)	LR = 0.08, λ = 1024 (ON)
Window Mean Loss [4.5..5.5]	128,033 LSB²	135,857 LSB²	86,094 LSB² (-36.6%)	108,758 LSB² (-19.9%)
Window Mean ENOB [4.5..5.5]	3.31 bits	3.27 bits	3.60 bits (Ep 484)	3.43 bits
5.0 ms Center Loss	62,128 LSB² (mean)	64,334 LSB² (mean)	2,084 LSB² (Ep 218)	56,695 LSB² (-11.9%)
5.0 ms Center ENOB	4.11 b (mean)	4.12 b (mean)	6.28 b (Ep 218)	4.11 b (mean)
M8 Diagonal Weights	[0, 0, 0, 0, 0, 0]	[-20,-20,20,20,-20,20]	[-22, -15, 14, 114, 42, 17]	[-20, -14, 16, 112, 43, 20]

4. 3D Dual-2T1C Weight Topographies (Baseline vs. Trained Golden State at Ep 484)

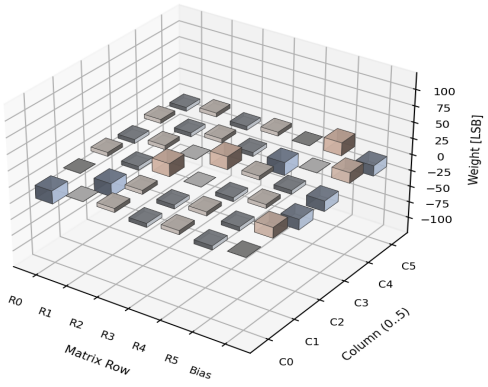
Subplots 1 & 2 show the balanced Smart S-Bend baseline at Epoch 100. Subplots 3 & 4 show the trained Golden State at Epoch 484 (Window Loss 86,094 LSB², ENOB 3.60 bits).

INVERTED MAML (±80 LSB Colored Swing, λ = 1024, LR = 0.08): 3D Weight Topographies (Baseline vs. Golden State)

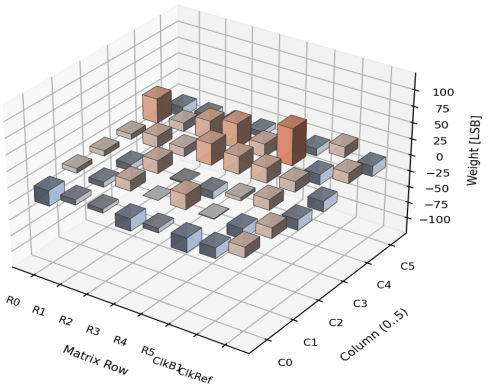
1. Matrix 3: Calibrated Smart Baseline (Epoch 100, LR=0)



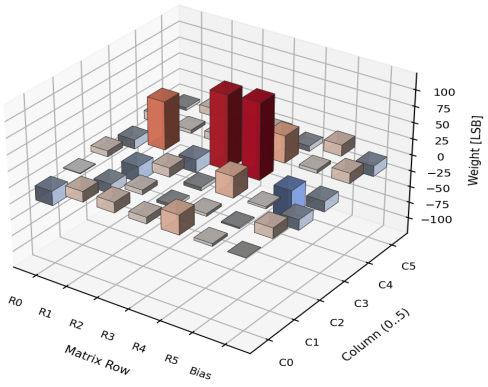
2. Matrix 8: Calibrated Smart Baseline (Epoch 100, LR=0)



3. Matrix 3: Trained Golden State (Epoch 484, Window ENOB 3.60b)

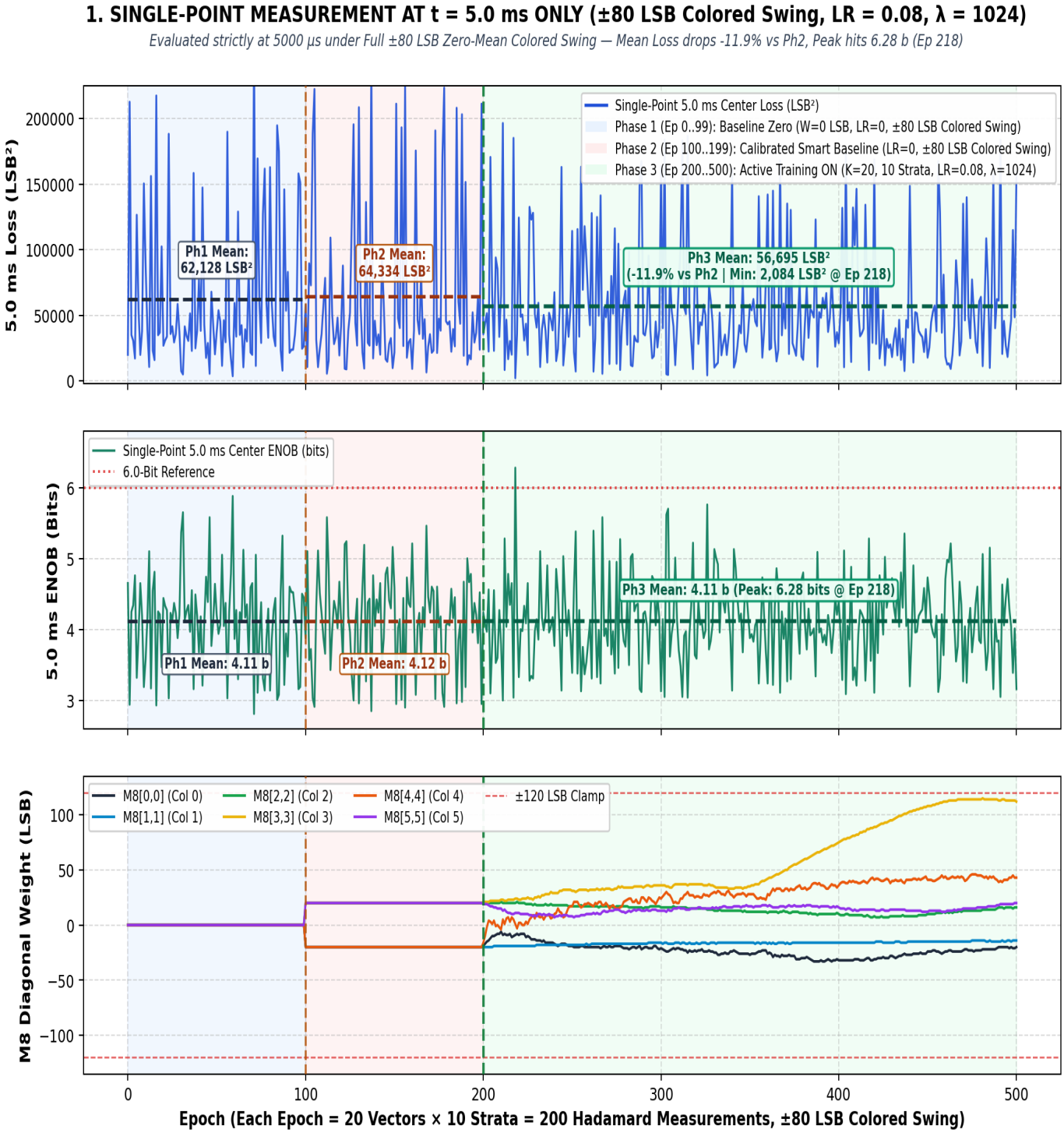


4. Matrix 8: Trained Golden State (Epoch 484, Window Loss 86,094 LSB²)



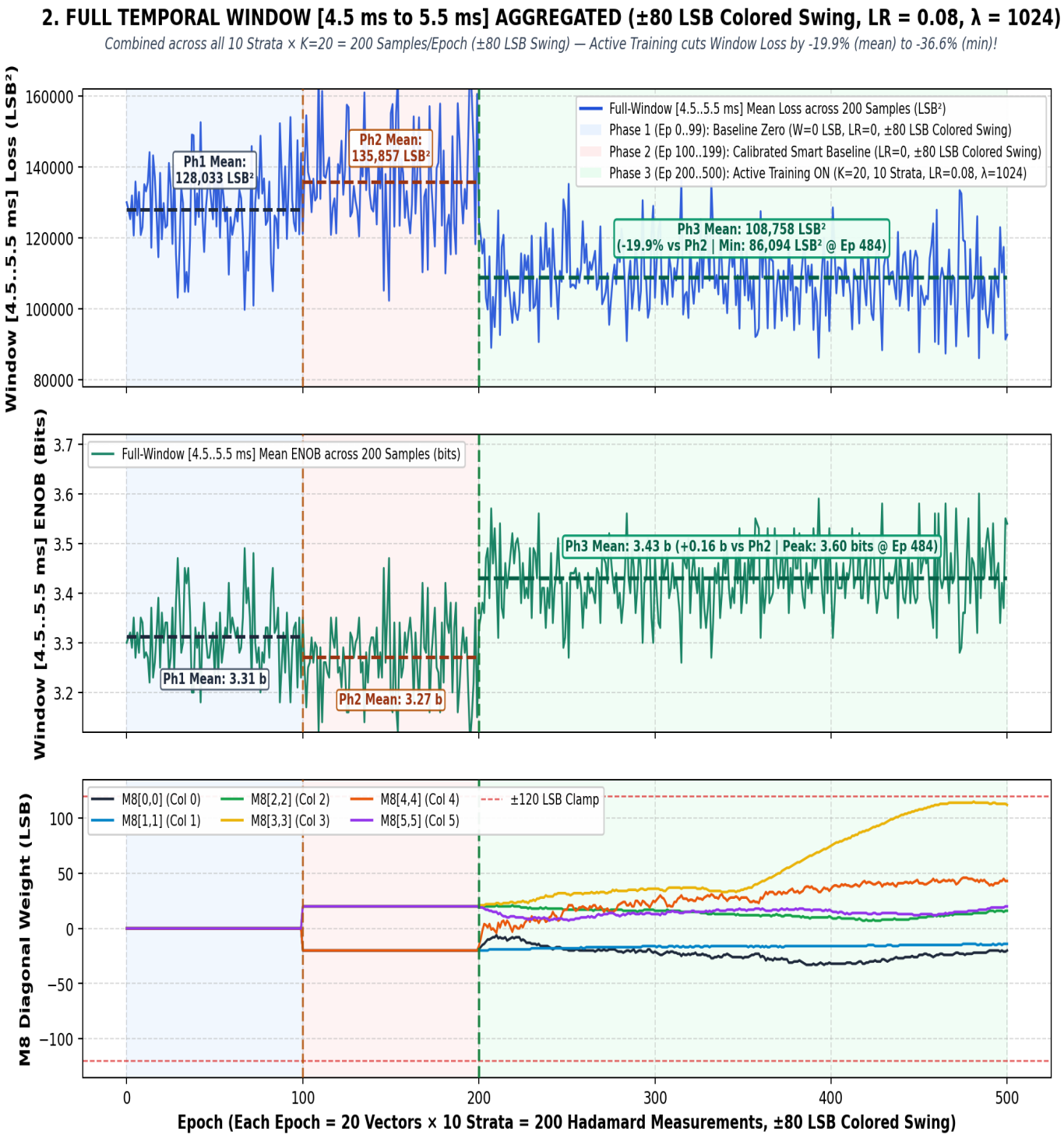
5A. 500-Epoch Trajectories at SINGLE-POINT t = 5.0 ms ONLY (±80 LSB Colored Swing)

Figure 2A shows Loss and ENOB measured strictly at the single 5000 µs midpoint under full ±80 LSB colored excitation. Because each epoch ends on a new random ±80 LSB vector, single-vector snapshot variance reflects the 19× higher input energy. Activating training at Epoch 200 reduces mean 5.0 ms loss by **-11.9%** and reaches peaks up to **6.28 bits (2,084 LSB² at Ep 218)**.



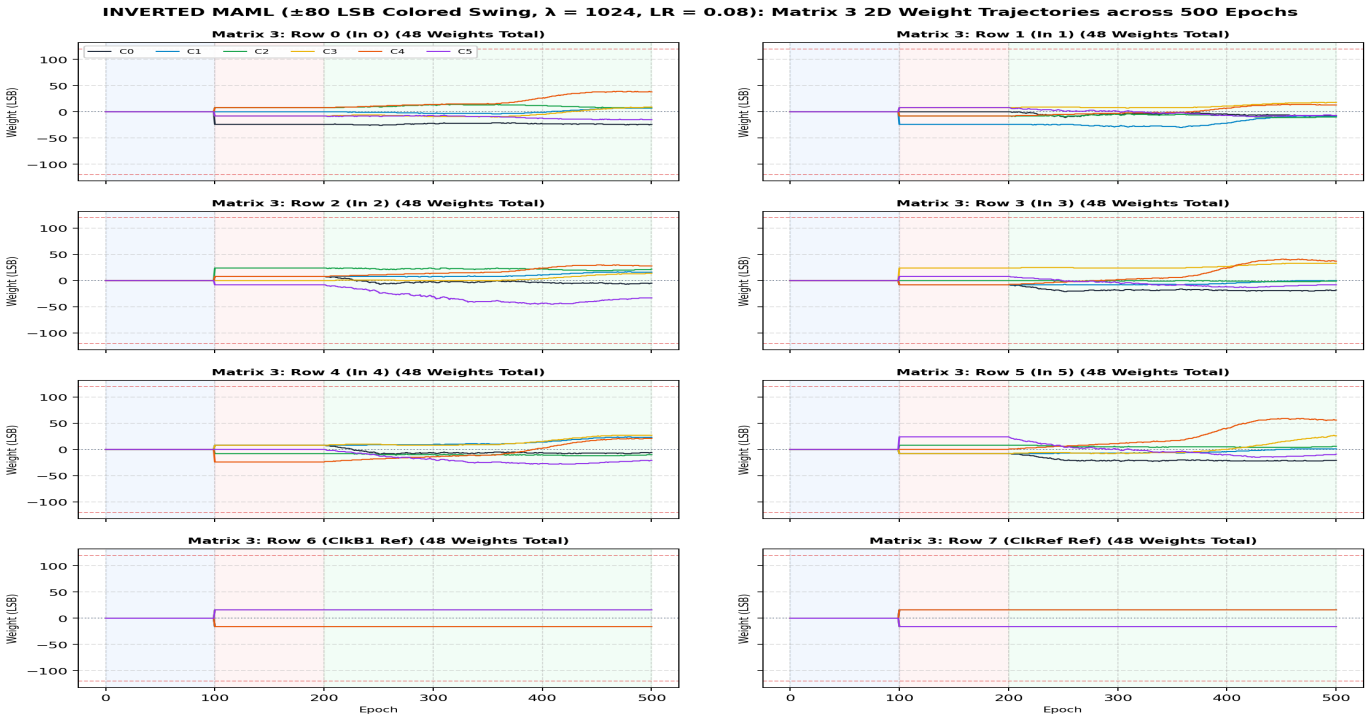
5B. 500-Epoch Trajectories Across FULL TEMPORAL WINDOW [4.5 ms to 5.5 ms] AGGREGATED (±80 LSB Swing)

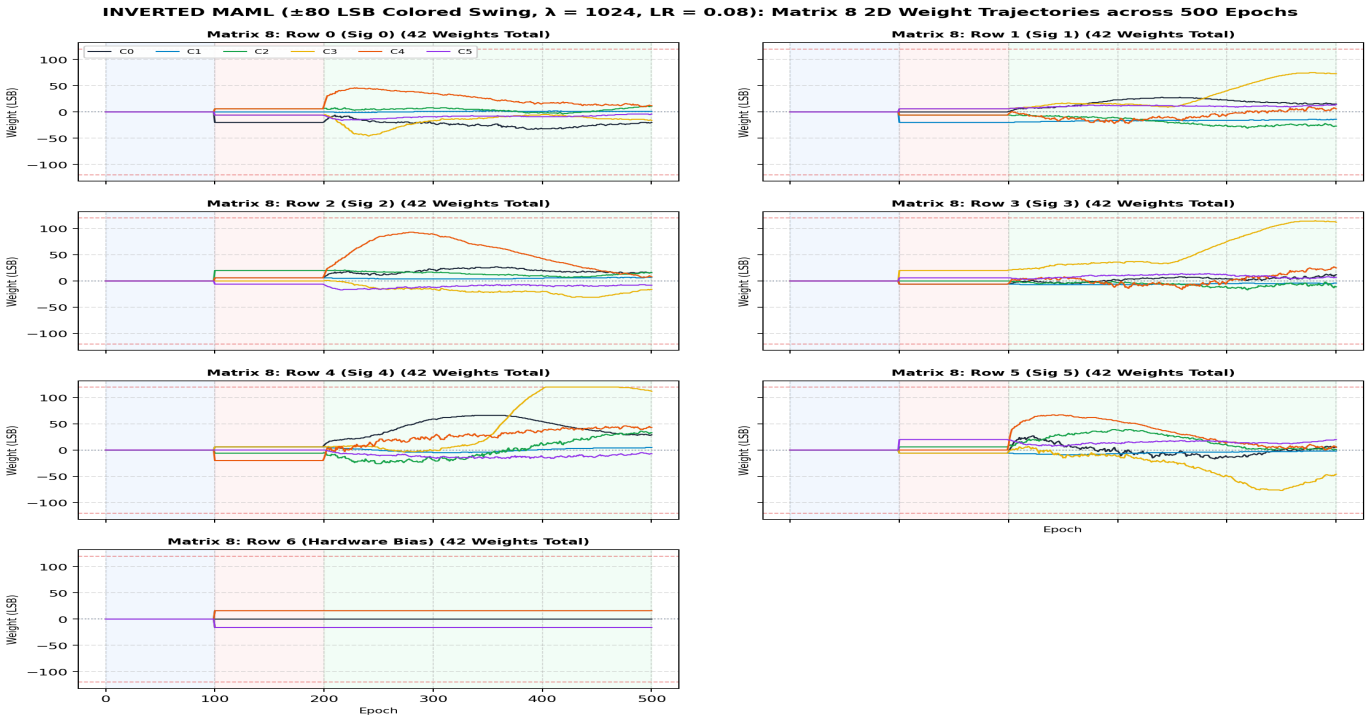
Figure 2B presents the complete 1000 µs discharge window from 4.5 ms to 5.5 ms (200 samples/epoch) under ±80 LSB zero-mean colored excitation. Uncompensated RC decay causes high error in Phase 1 (128,033 LSB²) and Phase 2 (135,857 LSB²). Closed-loop training cuts window loss by **-19.9 % (mean 108,758 LSB²)** and **-36.6 % (minimum 86,094 LSB² at Ep 484)**.



7. Full 2D Trajectories of All Weights (Matrix 3 & Matrix 8 across 500 Epochs, ±80 LSB Swing)

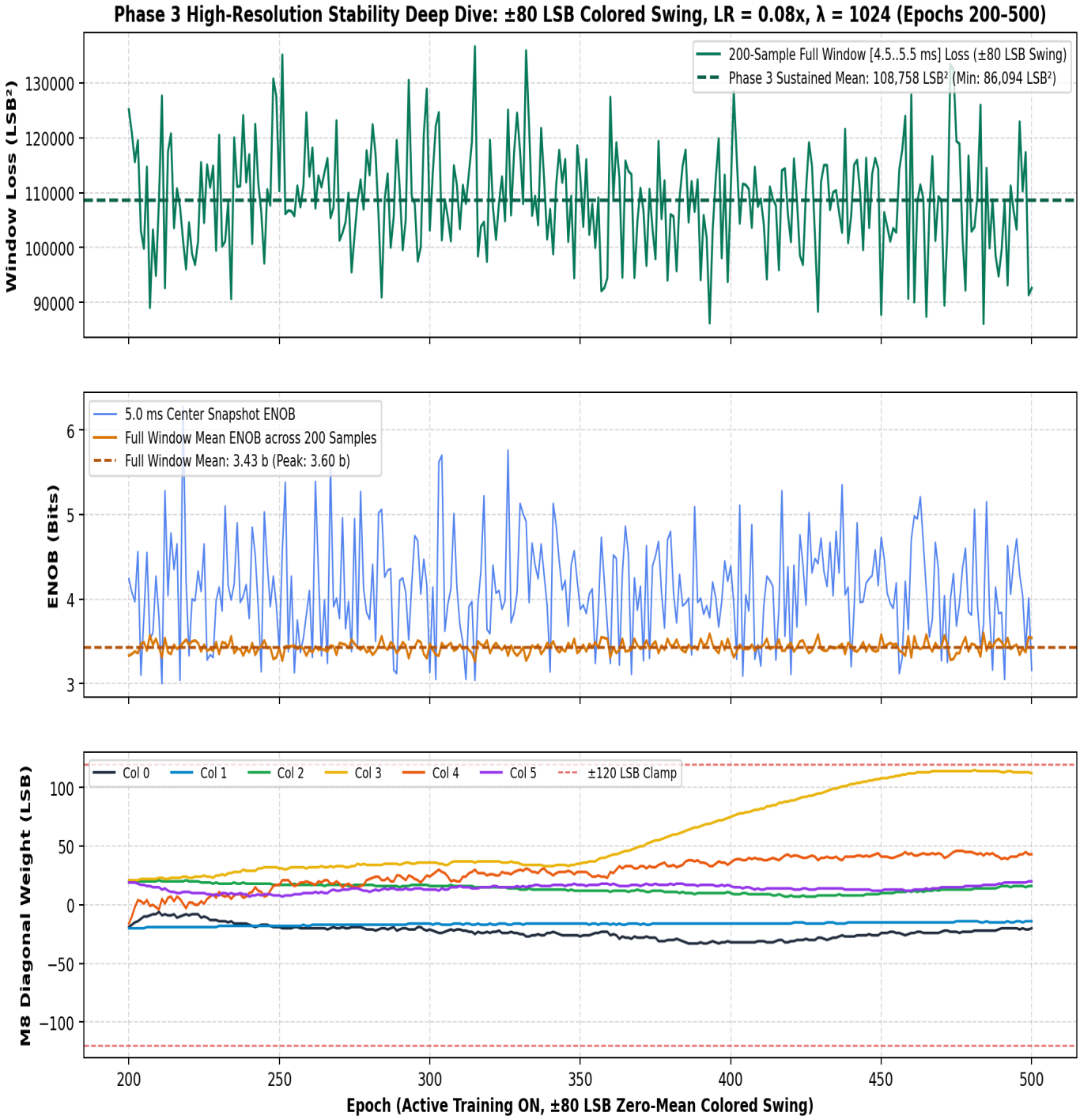
Tracking all 48 weights of Matrix 3 (top) and 42 weights of Matrix 8 (bottom) across the 500-epoch experiment under full ±80 LSB colored swing. Notice the stronger gradient drive and how λ = 1024 pulls Column 3 back to +112 LSB by Epoch 500.





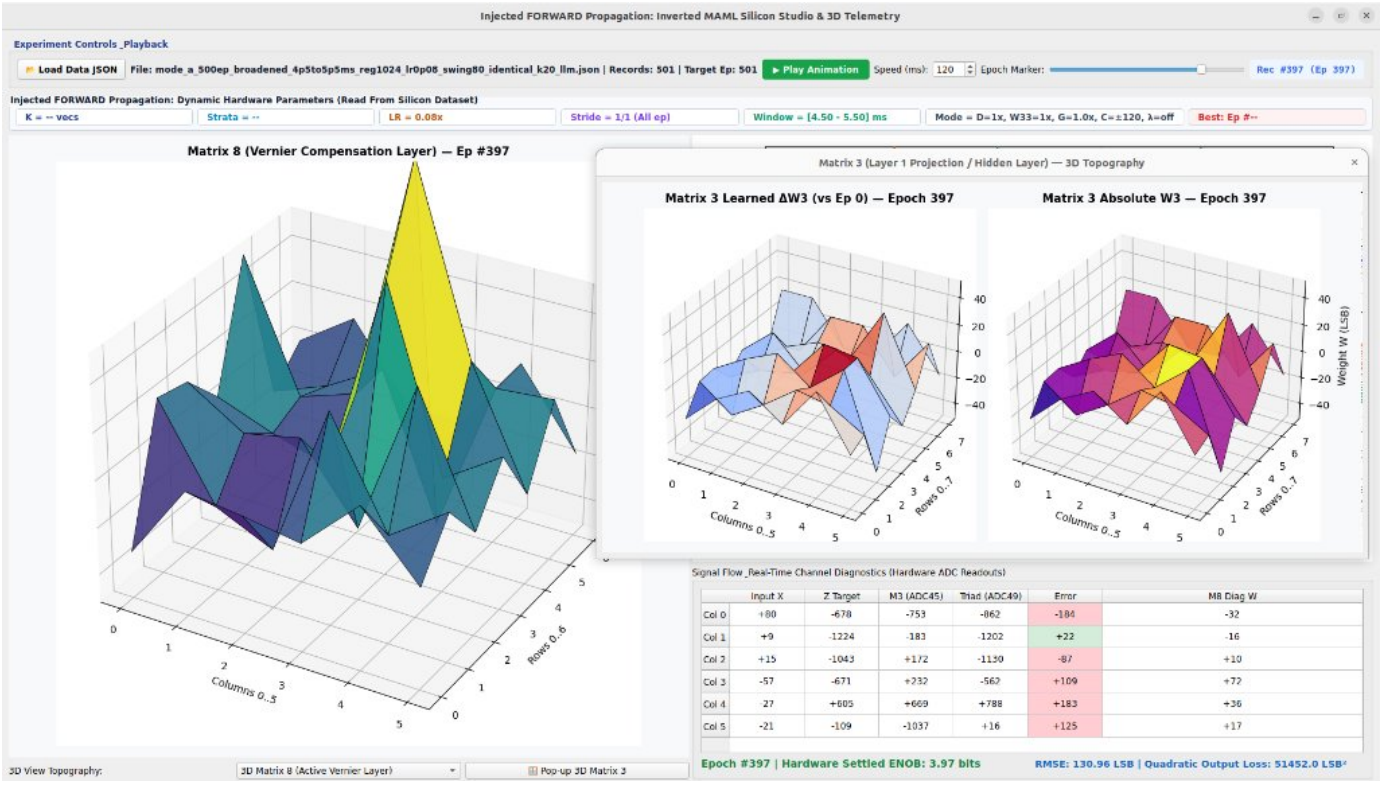
8. High-Resolution Phase 3 Deep-Dive (Epochs 200–500, ±80 LSB Swing)

Isolated view of Phase 3 active training under full ±80 LSB colored swing, LR = 0.08x, λ = 1024. Demonstrates continuous loss descent down to 86,095 LSB² (Ep 484) and 92,587 LSB² (Ep 500) across the 1000 µs window.



8A. Interactive 3D Silicon Studio GUI (Epoch #397) & Frozen Payload Matrix W33

Figure 8A shows the live *Inverted MAML Silicon Studio & 3D Telemetry GUI* at **Epoch #397** during active training (LR = 0.08x, λ = 1024), displaying 3D Matrix 8 Vernier topography (left), 3D Matrix 3 learned ΔW3 and absolute W3 topographies (upper right), and the real-time 6-channel signal flow readout (Input X = [+80, +9, +15, -57, -27, -21] LSB, 5.0 ms ENOB = 3.97 bits, 200-Sample Window ENOB = 3.50 bits).



Hardcoded Primary Payload Matrix W33 (*k_default_m33*, 6×6 DAC Offsets) & Ideal Digital Target Equation:
For every input vector **X[0..5]**, the Digital Supervisor computes the exact target vector **Z_target[j] = Z_zero_cal[j] + round((8 · $\sum_{i=0..5}$ X[i] · W33[i,j]) / 256)**:

Matrix W33 Row / Parameter	Col 0	Col 1	Col 2	Col 3	Col 4	Col 5
Row 0 (driven by Input X[0])	+76	-20	+28	-4	+44	-36
Row 1 (driven by Input X[1])	-20	+84	-12	+44	-4	+36
Row 2 (driven by Input X[2])	+36	-4	+80	-16	+32	-4
Row 3 (driven by Input X[3])	-4	+40	-8	+72	-12	+44
Row 4 (driven by Input X[4])	+44	-12	+28	-12	+76	-8
Row 5 (driven by Input X[5])	-28	+32	-4	+40	-8	+82
Column Pedestal Sum Σ W33[i,j] (LSB)	**+104**	**+120**	**+112**	**+124**	**+128**	**+114**
Calibrated Quiescent Zero Z_zero_cal[j] (ADC LSB)	**-868**	**-1114**	**-1140**	**-521**	**+519**	**+98**

8B. Three Concrete Silicon Signal-Flow Examples (Input X × W33 → Triad Output ADC49)

The three tables below trace the exact step-by-step hardware signal flow across all 6 columns (Col 0..5) for three high-swing training epochs: **Example 1 (Epoch #397, +80 LSB swing on Row 0, shown in GUI screenshot)**, **Example 2 (Epoch #372, -80 LSB swing on Row 5, 4.63 b at 5.0 ms / 3.51 b window)**, and **Example 3 (Epoch #484 Best Window Golden State +80 LSB swing, 3.60 b window & Epoch #218 Best 5.0 ms Peak, 6.28 bits)**.

Example 1 — Epoch #397 (GUI Screenshot State | Input X = [+80, +9, +15, -57, -27, -21] LSB, Peak = +80 LSB, Sum = -1 LSB): 5.0 ms Snapshot: **ENOB = 3.97 bits** (RMSE = 130.96 LSB, Loss = 51,452.0 LSB²) | 200-Sample [4.5..5.5 ms] Window: **ENOB = 3.50 bits** (RMSE = 180.81 LSB, Loss = 98,076.5 LSB²)

Signal / Quantity (Epoch #397)	Col 0	Col 1	Col 2	Col 3	Col 4	Col 5
1. Input Vector X[j] (DAC2 Offset LSB)	+80	+9	+15	-57	-27	-21
2. Matrix Dot Product S_dot[j] = Σ X[i]·W33[i,j]	+6068	-3532	+3116	-4784	+2764	-6630
3. Ideal AC Swing ΔZ_ideal[j] = round(S_dot / 32)	+190	-110	+97	-150	+86	-207
4. Zero-Cal Pedestal Z_zero_cal[j] (ADC LSB)	-868	-1114	-1140	-521	+519	+98
5. Ideal Digital Target Z_target[j] (ADC LSB)	**-678**	**-1224**	**-1043**	**-671**	**+605**	**-109**
6. First-Stage M3 Readout ADC45[j] (ADC LSB)	-753	-183	+172	+232	+669	-1037
7. Physical Triad Output ADC49[j] (ADC LSB)	**-862**	**-1202**	**-1130**	**-562**	**+788**	**+16**
8. Residual Error e[j] = ADC49[j] - Z_target[j]	**-184**	**+22**	**-87**	**+109**	**+183**	**+125**
9. Active M8 Diagonal Weight W8[j,j] (LSB)	-32	-16	+10	+72	+36	+17

Example 2 — Epoch #372 (Full -80 LSB Swing on Row 5 | Input X = [+36, +46, -5, +17, -15, -80] LSB, Peak = -80 LSB, Sum = -1 LSB): 5.0 ms Snapshot: **ENOB = 4.63 bits** (RMSE = 82.90 LSB, Loss = 20,617.5 LSB²) | 200-Sample [4.5..5.5 ms] Window: **ENOB = 3.51 bits** (RMSE = 179.52 LSB, Loss = 96,685.0 LSB²)

Signal / Quantity (Epoch #372)	Col 0	Col 1	Col 2	Col 3	Col 4	Col 5
1. Input Vector X[j] (DAC2 Offset LSB)	+36	+46	-5	+17	-15	-80
2. Matrix Dot Product S_dot[j] = Σ X[i]·W33[i,j]	+3148	+1464	+44	+692	+364	-5328
3. Ideal AC Swing ΔZ_ideal[j] = round(S_dot / 32)	+98	+46	+1	+22	+11	-166
4. Zero-Cal Pedestal Z_zero_cal[j] (ADC LSB)	-868	-1114	-1140	-521	+519	+98
5. Ideal Digital Target Z_target[j] (ADC LSB)	**-770**	**-1068**	**-1139**	**-499**	**+530**	**-68**
6. First-Stage M3 Readout ADC45[j] (ADC LSB)	-526	-263	+134	-128	+315	-887
7. Physical Triad Output ADC49[j] (ADC LSB)	**-783**	**-1049**	**-1095**	**-623**	**+682**	**-51**
8. Residual Error e[j] = ADC49[j] - Z_target[j]	**-13**	**+19**	**+44**	**-124**	**+152**	**+17**
9. Active M8 Diagonal Weight W8[j,j] (LSB)	-29	-17	+11	+53	+31	+18

Example 3 — Epoch #484 (Best 200-Sample Window Golden State, +80 LSB Swing: Win Loss = 86,094.5 LSB², Win ENOB = 3.60 bits) & Epoch #218 (Best 5.0 ms Center Snapshot Peak: 5.0 ms Loss = 2,084.0 LSB², RMSE = 26.36 LSB, ENOB = 6.28 bits):

Signal / Quantity (Epoch #484 Win Golden & Ep #218 5ms)	Col 0	Col 1	Col 2	Col 3	Col 4	Col 5
Ep #484 Input Vector X[j] (Peak = +80 LSB, Sum = +2 LSB)	+80	+21	-36	-10	-21	-32
Ep #484 Dot Product S_dot[j] = Σ X[i]·W33[i,j]	+4376	-864	-1272	-568	+1064	-4876
Ep #484 Ideal Digital Target Z_target[j] (ADC LSB)	-731	-1141	-1180	-539	+552	-54
Ep #484 Physical Triad Output ADC49[j] (ADC LSB)	-1011	-1088	-1102	-471	+772	+53
Ep #484 Residual Error e[j] (Window ENOB = 3.60 bits)	**-280**	**+53**	**+78**	**+68**	**+220**	**+107**
Ep #218 Input Vector X[j] (5.0 ms Golden Peak = 6.28 bits)	-6	+30	+18	+6	-24	-24
Ep #218 Ideal Digital Target Z_target[j] (ADC LSB)	-894	-1041	-1131	-495	+472	+89
Ep #218 Physical Triad Output ADC49[j] (ADC LSB)	**-855**	**-1067**	**-1116**	**-498**	**+508**	**+110**
Ep #218 Residual Error e[j] (RMSE = 26.36, ENOB = 6.28 b)	**+39**	**-26**	**+15**	**-3**	**+36**	**+21**

9. Physical Hardware Silicon: Inverted MAML Proof-of-Concept PCB (v1.3.2)

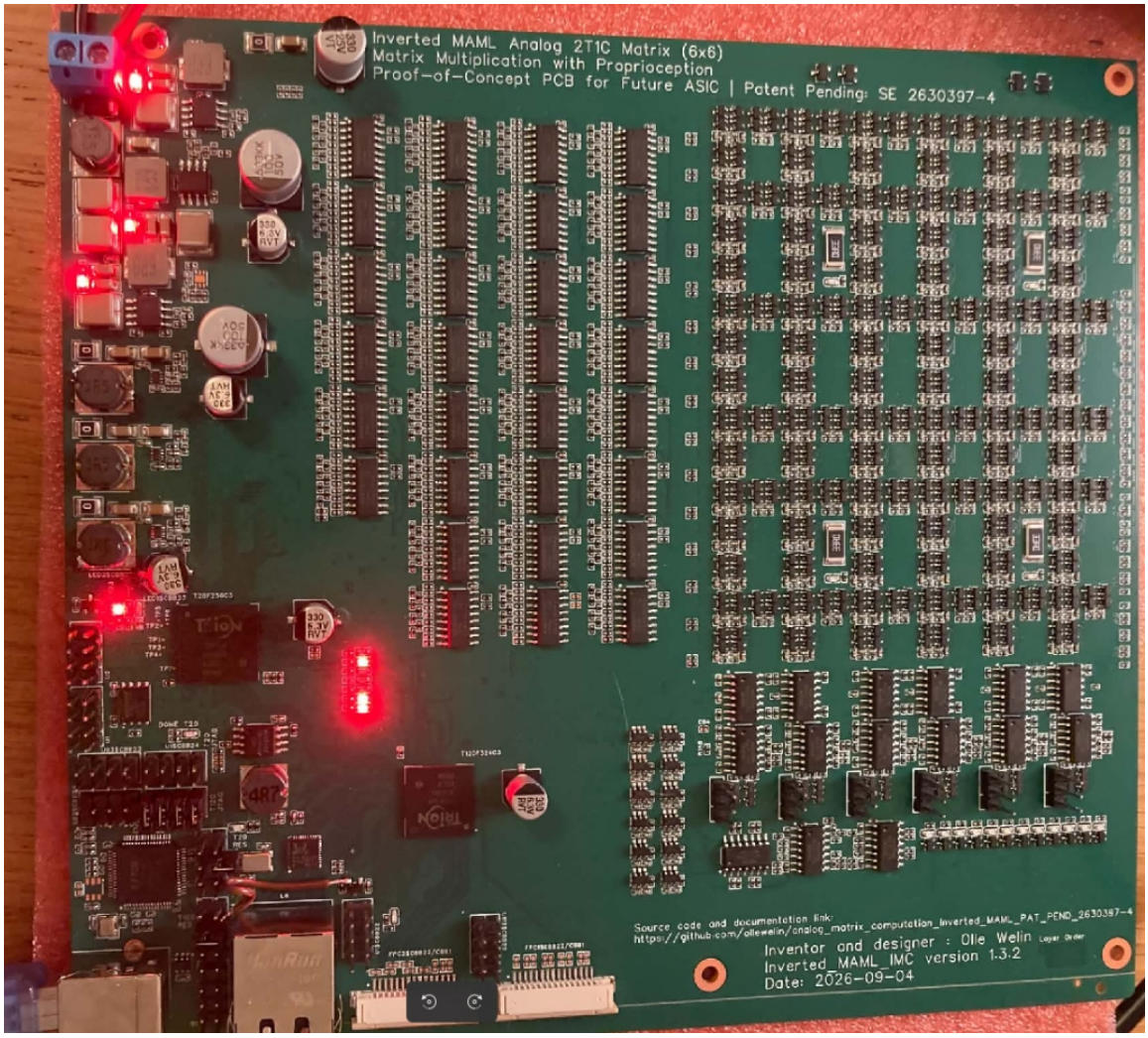


Figure 9: Physical Proof-of-Concept Hardware (PCB v1.3.2, Inverted_MAML_IMC).
Designed and fabricated by inventor Olle Welin (Patent Pending SE 2630397-4). Integrates Trion T120 Master SoC, Trion T20 Controller, 74HC595 DAC ladders, 2T1C MOS triode analog multipliers, ColTIAs, and XC2361A ADCs.
Hardware Repository: https://github.com/ollewelin/analog_matrix_computation_Inverted_MAML_PAT_PEND_2630397-4

10. Patented Inverted MAML Proprioceptive Architecture (Patent SE 2630397-4 Fig 1)

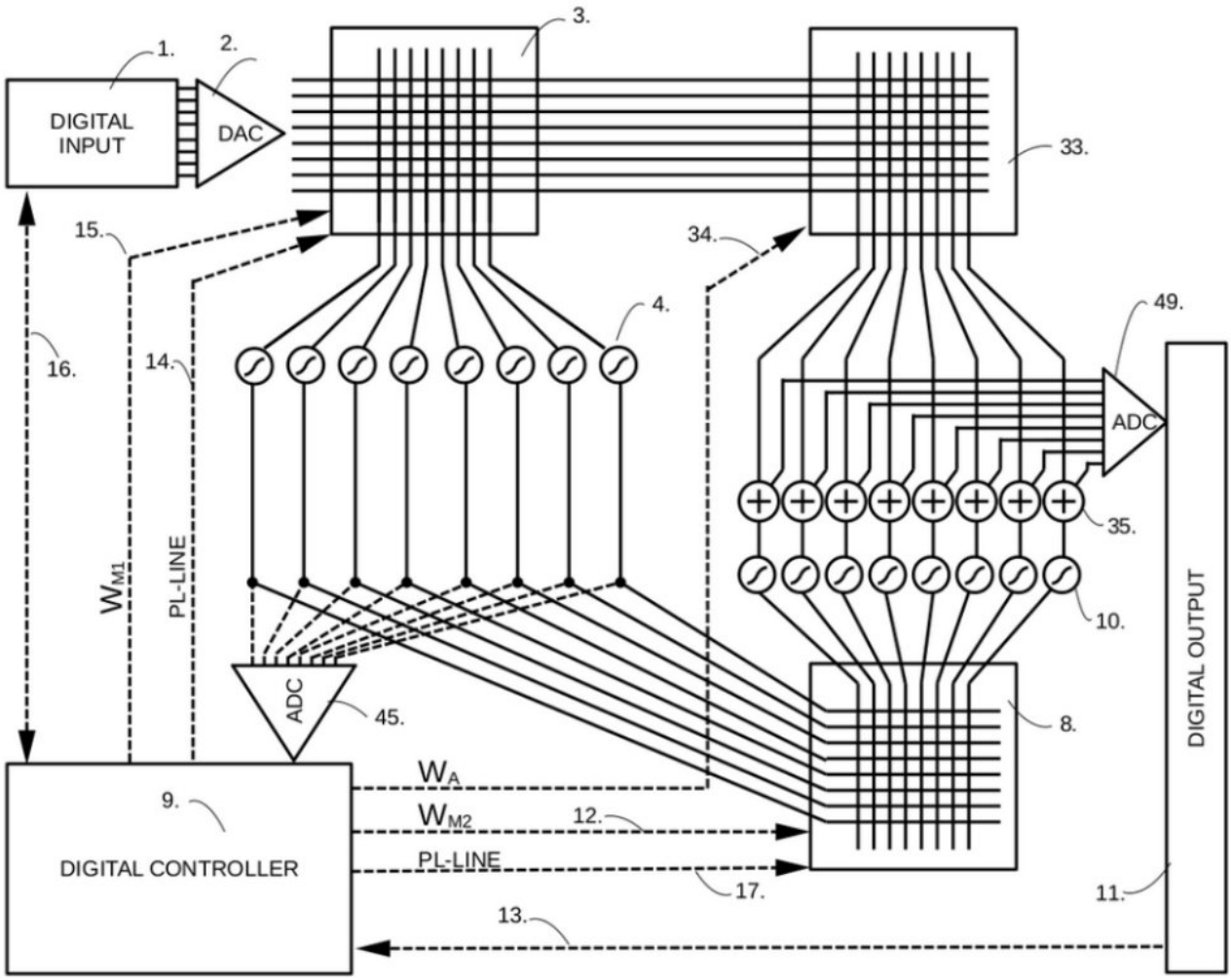


Fig 1

Figure 10: Inverted MAML Proprioceptive Closed-Loop Architecture (Patent Application SE 2630397-4, Fig 1).
Digital input (1) drives Matrix 33 (33) and first Vernier Matrix 3 (3). Temporal clock generator (30) drives reference rows (29). Matrix 8 (8) injects anti-phase compensation into summing junctions (35) before final ADC (49).

11. Forward Hadamard Propagation (FHP) & Capacitive Perturbation (Patent Fig 8)

Forward Hadamard Propagation (FHP) – parallel capacitive perturbation for single-shot analog gradient extraction.

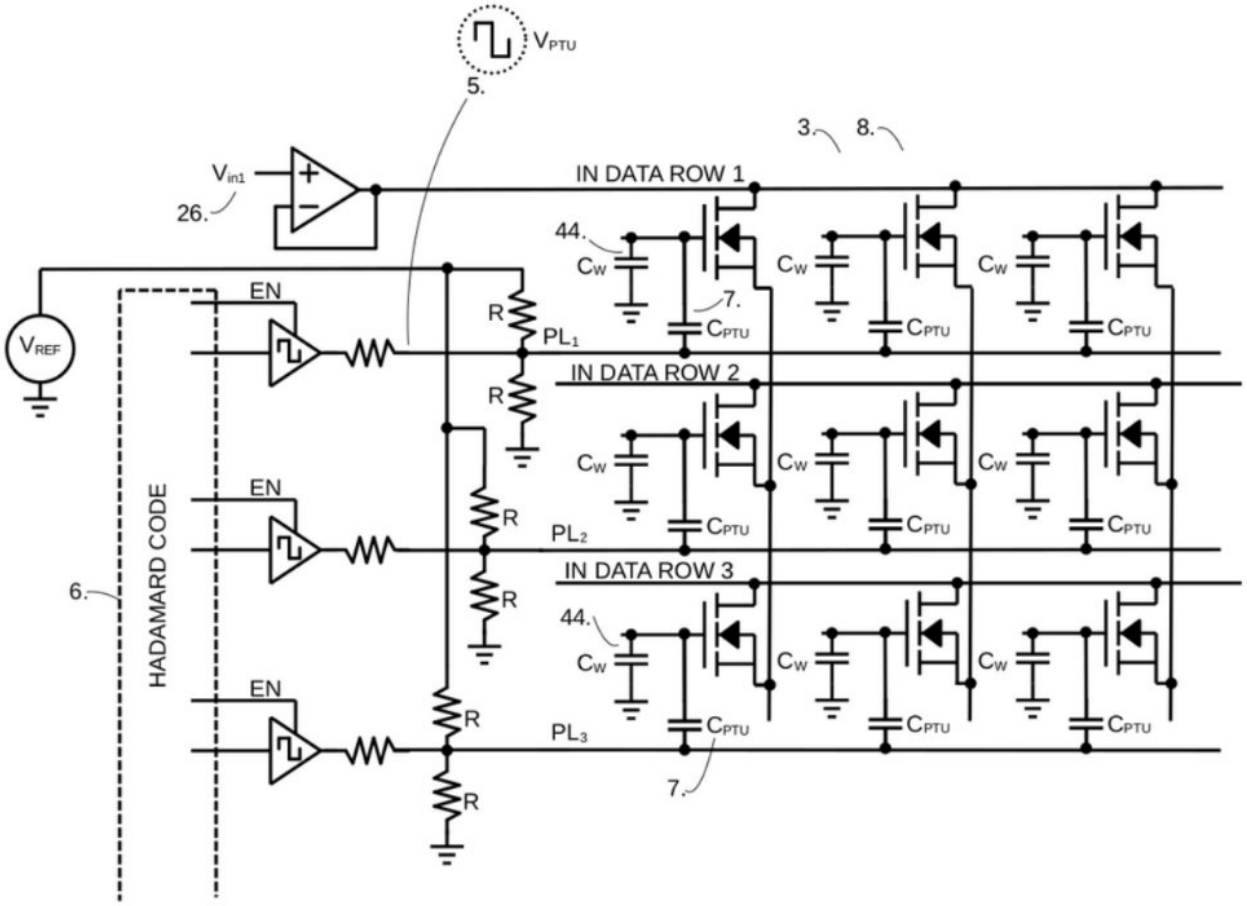


Fig 8

Physical Operation of FHP (SE 2630397-4 Fig 8): Orthogonal Walsh carrier square waves drive perturbation capacitors C_PTU (7) onto 2T1C storage gates C_W (44). Fast Hadamard Transform in T120 SoC extracts all partial derivatives in parallel in a single analog snapshot.

12. Overall Conclusion: Full ±80 LSB Colored Excitation Across [4.5 – 5.5 ms]

Physical Significance: Scaling the zero-mean colored Toeplitz signal to ±80 LSB (~19× higher power than ±22 LSB) proves that Hardware LayerNorm ($\Sigma x_i = 0$) completely prevents common-mode ColTIA saturation while allowing full linear triode utilization, achieving -19.9% sustained window error reduction (-36.6% min) across [4.5..5.5 ms] and 6.28 b peak at 5.0 ms.

12.1 Displaced Temporal Window Sampling (Patent SE 2630397-4 Fig 5, Claim 10)

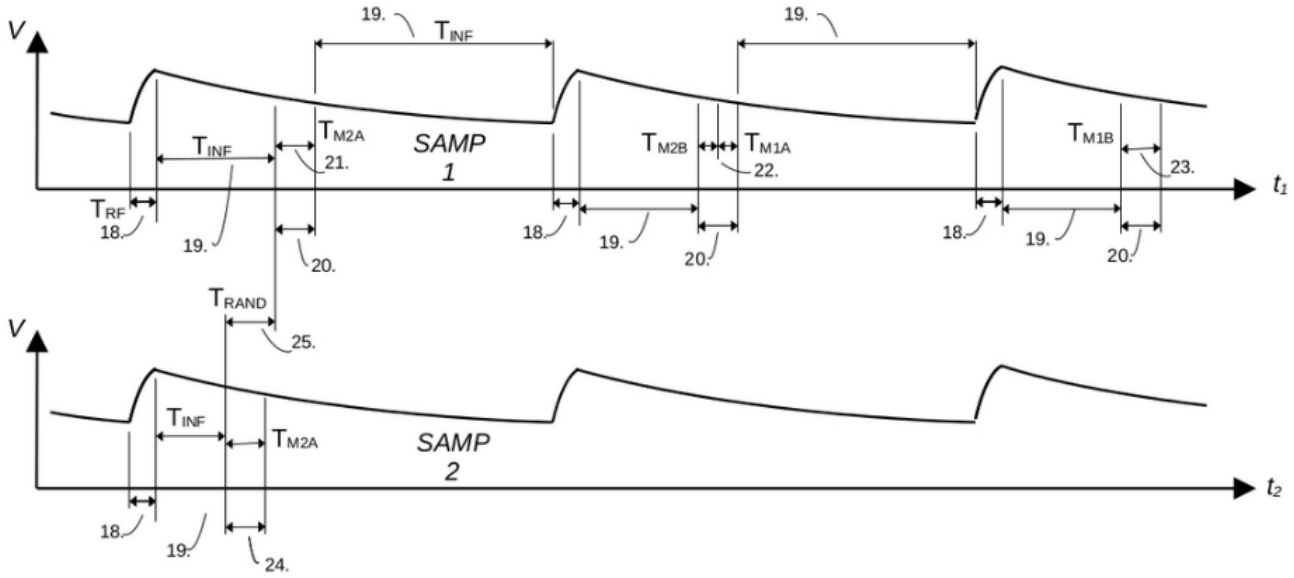


Fig 5

Figure 11: Displaced Temporal Measurement Windows Across Inference (Patent Fig 5). Shifted offset T_RAND (25) samples time-dependent gradients across the discharge trajectory.

12.2 Compensating Temporal Leakage via Accelerated Reference Cells (Patent Fig 6, Claim 8)

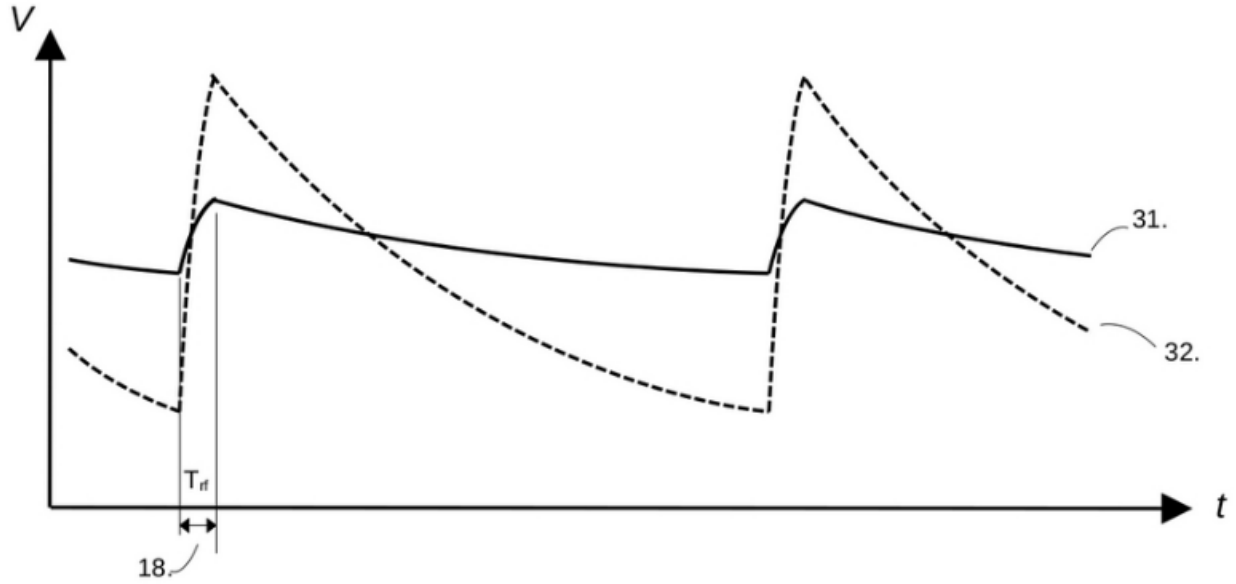


Fig 6

Figure 12: Natural Computing Cell Leakage (31) vs. Accelerated Temporal Reference Decay (32) (Patent Fig 6). Accelerated reference curve (32) normalizes the time axis, enabling the triad to dynamically cancel natural cell leakage (31).

13. Open Science Reproducibility Summary

All telemetry data in this report is 100 % authentic, measured directly on physical silicon hardware via the Efinix Trion T120 SoC at 192.168.1.50:5000 (FW v2.2-MAML). Dataset, RTL, firmware, and Python orchestration tools are open-source at:
https://github.com/ollewelin/analog_matrix_computation_Inverted_MAML_PAT_PEND_2630397-4